

Reinforcement Learning for Rocket Landing Control

A Practical Research Topic on Learning-Based Control of an Unstable, Nonlinear System

Topic 3 • AI & Control Theory¹

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Abstract

Rocket landing is a challenging control problem in which a rocket must land safely while managing fuel, speed, position, angle, and engine thrust. The system is unstable, nonlinear, and affected by gravity and motion constraints. Traditional control uses physics-based models and mathematical controllers; reinforcement learning offers an alternative in which an agent learns a landing policy through repeated trials in a simulated environment. This document proposes a practical research project that investigates whether a reinforcement learning agent can learn a stable and fuel-efficient rocket landing policy in a simulated environment, and outlines the background, the core research question and sub-questions, a five-step research plan, the expected outcome, and source references.

1. Basic Background

Rocket landing is a challenging control problem. A rocket must land safely while managing fuel, speed, position, angle, and engine thrust. The system is difficult because the rocket is unstable, nonlinear, and affected by gravity and motion constraints.

Traditional control methods use physics-based models and mathematical controllers. Reinforcement learning provides another approach: an agent learns how to control the rocket by repeatedly trying to land in a simulated environment. The agent observes the rocket's state, chooses an action, receives a reward or penalty, and gradually learns a better landing policy.

This topic is more AI-related than the first two, but it is still strongly connected to algorithms, control theory, simulation, optimization, and sequential decision-making.

2. Research Question

Can a reinforcement learning agent learn a stable and fuel-efficient rocket landing policy in a simulated environment?

Possible sub-questions

1. Which reward design leads to better landing performance?
2. Can the agent balance landing accuracy and fuel efficiency?
3. How robust is the learned policy under wind, sensor noise, or different starting positions?
4. How does reinforcement learning compare with a simple rule-based controller?

3. Brief Research Plan

1. Create or use a rocket landing simulator

- Start with a 2D simulation.
- State variables may include position, velocity, angle, angular velocity, and fuel.
- Actions may include main engine thrust and side thrusters.

2. Define the reward function

- Positive reward for safe landing.
- Penalty for crashing.
- Penalty for high landing speed.
- Penalty for using too much fuel.
- Penalty for large tilt angle near landing.

3. Train reinforcement learning models

- Start with a standard RL algorithm such as Q-learning for simple discrete control or PPO/DQN for more advanced simulation.
- Train over many simulated landing attempts.

4. Compare with baseline controllers

- Random controller.
- Rule-based controller.
- Proportional control or simple physics-based controller.

5. Evaluate performance

- Landing success rate.
- Average fuel consumption.
- Final landing speed.
- Final landing distance from target.
- Robustness under different initial conditions.

4. Expected Outcome

The project should show how reinforcement learning can learn a control strategy through trial and error. A strong version of the project would compare different reward functions and show how reward design affects safety, fuel efficiency, and stability.

References

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